

## **Cardiac Arrhythmia Classification**

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## Abstract

Irregular heart rhythm disorders represent a paramount driver of adverse cardiovascular events worldwide. Continuous monitoring via single-lead (Lead-I) wearable electronics presents an unprecedented screening opportunity outside clinical facilities. However, building machine learning models for single-lead signals requires addressing severe class skew between normal and abnormal heartbeats, heavy ambulatory noise, and potential accuracy drops when evaluating unseen patients. Our team designed an end-to-end multi-label machine learning framework evaluating 21,799 10-second single-lead ECG recordings from 18,869 patients within the PhysioNet PTB-XL database. Operating on 100 Hz low-resolution signals (`_lr.dat` files containing 1,000 voltage samples per record), our workflow executes signal hygiene with median demographic imputation, extracts 45 timing, wave morphology, and power-spectral features, and trains cost-weighted XGBoost models across seven cardiac conditions: Normal Sinus Rhythm, Atrial Fibrillation/Flutter, First-Degree AV Block, Premature Atrial Contraction, Premature Ventricular Contraction, Bradycardia, and Tachycardia.

Evaluation on held-out test data (Fold 10,  $n = 2,195$ ) established that XGBoost achieved a mean ROC-AUC of 0.961, outperforming baseline decision trees (0.922 AUC). Decision threshold tuning via Youden's J statistic boosted sensitivity for First-Degree AV Block (+57.0% to 88.6%) and Premature Atrial Contraction (+47.5% to 92.5%). Sigmoid calibration aligned probability outputs, while noise stress-testing identified 10 dB SNR as a resilient operational limit (0.806 AUC). Patient-stratified splitting restricted inter-patient accuracy decay to under 0.023 AUC, demonstrating strong clinical viability. Project resources are accessible at [https://github.com/abhairwal-usd/intro\\_ai\\_ml\\_aai501](https://github.com/abhairwal-usd/intro_ai_ml_aai501).

# Cardiac Arrhythmia Classification

## 1. Introduction

### 1.1 Business Context and Clinical Objectives

Early identification of cardiac arrhythmias is vital for preventing life-threatening conditions such as ischemic stroke and heart failure. Standard hospital diagnostic procedures rely on 12-lead electrocardiograms (ECGs) to capture comprehensive spatial electrical vectors. While 12-lead ECGs represent the medical gold standard, they require physical visits, specialized equipment, and trained personnel, capturing only brief diagnostic snapshots.

Consumer wearable devices (smartwatches, ambulatory patches) enable continuous single-lead (Lead-I) tracking in free-living environments, recording potential differences between the left and right arms. Implementing automated machine learning classifiers for Lead-I signals requires overcoming significant obstacles: missing spatial vectors, severe imbalance between healthy and pathological heartbeats, motion artifacts, and patient-level overfitting. Source code and scripts supporting our system are hosted at [https://github.com/abhairwal-usd/intro\\_ai\\_ml\\_aai501](https://github.com/abhairwal-usd/intro_ai_ml_aai501).

### 1.2 Anchor Paper Alignment and Research Questions

Our research design builds upon the single-lead benchmark established by Sadiq et al. (2025) in the IEEE Journal of Biomedical and Health Informatics. We address four core research questions:

1. Research Question 1: Can 45 signal-derived morphological, timing, and spectral features extracted from Lead-I signals achieve robust multi-label discrimination across seven cardiac rhythm conditions?
2. Research Question 2: How effectively do cost-sensitive positive-class loss weighting and Youden’s J threshold tuning resolve sensitivity tradeoffs in rare arrhythmia types?

3. Research Question 3: What is the quantitative signal-to-noise ratio (SNR) boundary at which environmental noise degrades diagnostic accuracy below acceptable screening standards?
4. Research Question 4: Does patient-stratified data splitting prevent inter-patient performance decay across unseen individuals, and is diagnostic parity maintained across age and sex cohorts?

### 1.3 Scope and Target Arrhythmia Definitions

Following the diagnostic taxonomy of Sadiq et al. (2025), our project classifies seven clinical cardiac rhythm categories in a multi-label binary setting. Table 1 defines the clinical scope and diagnostic characteristics of these seven target classes.

**Table 1**  
*Target Arrhythmia Labels and Clinical Definitions*

| Label     | Condition Name                    | Clinical Description                                 | Diagnostic Sign on Lead-I                         |
|-----------|-----------------------------------|------------------------------------------------------|---------------------------------------------------|
| NSR       | Normal Sinus Rhythm               | Normal cardiac electrical activity, rate 60–100 bpm. | Regular P-QRS-T complexes.                        |
| AFb / AFl | Atrial Fibrillation / Flutter     | Chaotic or rapid atrial activity; high stroke risk.  | Absent P-waves, irregular RR intervals.           |
| 1AV       | First-Degree AV Block             | Delayed conduction through the AV node.              | PR interval > 200 ms (prolonged).                 |
| PAC       | Premature Atrial Contraction      | Early heartbeat originating in the atria.            | Abnormal or early P-wave, non-compensatory pause. |
| PVC       | Premature Ventricular Contraction | Early heartbeat originating in ventricles.           | Wide, bizarre QRS complex (>120 ms).              |
| BRADY     | Bradycardia                       | Abnormally slow heart rate (<60 bpm).                | Prolonged RR intervals (>1000 ms).                |
| TACHY     | Tachycardia                       | Abnormally fast heart rate (>100 bpm).               | Shortened RR intervals (<600 ms).                 |

*Note.* Descriptions adapted from PTB-XL standards and clinical arrhythmia classification benchmarks.

## 2. Related Work

### 2.1 ECG and Cardiac Arrhythmia Classification in Literature

Automated ECG analysis expanded from early rule-based QRS detectors like the Pan-Tompkins peak algorithm (Pan & Tompkins, 1985) to machine learning methodologies. Statistical classifiers operating on timing and morphological heartbeat features demonstrated strong performance on benchmark datasets (De Chazal et al., 2004). Deep learning architectures later enabled automated end-to-end feature extraction from raw signal arrays.

### 2.2 Single-Lead vs. 12-Lead Wearable Diagnostics

Deploying deep models trained on clinical 12-lead ECGs to single-lead wearable hardware introduces operational challenges. Sadiq et al. (2025) highlighted that models trained on pristine clinical recordings degrade when exposed to single-lead ambulatory signals corrupted by motion artifacts. Rigorous validation also requires patient-isolated data splitting to avoid patient memorization.

### 2.3 Integration of Team Project System and Empirical Diagnostics

Our team’s unified system combines custom feature engineering, dynamic cost-sensitive loss weighting, gradient boosted tree ensembles, decision threshold tuning, and probability calibration. Rather than relying on synthetic resampling techniques like SMOTE, which can alter multi-label correlations, our workflow computes exact label-specific loss weights from positive class prevalence. Tuning decision boundaries using Youden’s J statistic and stress-testing under simulated Gaussian noise establishes a lightweight, resilient screening solution.

## 3. Method

### 3.1 PTB-XL Dataset Profile and Demographics

Our analysis evaluates the PhysioNet PTB-XL ECG repository v1.0.3 (Wagner et al., 2020), containing 21,799 10-second 12-lead ECG recordings from 18,869 patients. We extract Lead-I voltage signals sampled at 100 Hz (`_lr.dat` files, 1,000 voltage samples per record) to replicate wearable hardware parameters. Table 2 details the dataset demographics.

**Table 2**  
*PTB-XL Raw Dataset Profile and Demographics*

| Dataset Metric               | Raw PTB-XL Value  | Cleaned Project Value | Notes and Filtering Impact                       |
|------------------------------|-------------------|-----------------------|--------------------------------------------------|
| Total ECG Recordings ( $n$ ) | 21,799            | 21,786                | 13 missing or corrupted records removed.         |
| Distinct Patients ( $n$ )    | 18,869            | 18,862                | Multi-recording patients grouped by ID.          |
| Signal Duration              | 10 seconds        | 10 seconds            | 1,000 continuous Lead-I voltage samples.         |
| Sampling Rate                | 100 Hz / 500 Hz   | 100 Hz                | Low-resolution standard for wearable efficiency. |
| Age (Years, Mean $\pm$ SD)   | 59.8 $\pm$ 16.9   | 59.8 $\pm$ 16.8       | Filtered extreme outliers (age > 100).           |
| Sex Distribution             | 52.1% M / 47.9% F | 52.1% M / 47.9% F     | Balanced demographic representation.             |
| Quality Contamination        | 30.33% noisy      | 30.33% flagged        | Preserved for noise degradation stress-testing.  |

*Note.* Dataset profile derived from PTB-XL v1.0.3 low-resolution 100 Hz Lead-I records (1,000 voltage samples per 10-second signal).

### 3.2 Data Preparation, Imputation, and Quality Indicators

Data preparation established signal hygiene under the CARE-I framework. Missing height and weight values were imputed using median values stratified by age decile and sex. Quality columns were converted into binary flags tracking baseline wander, static noise, burst noise, extra beats, pacemaker spikes, and electrode contact issues. Table 3 details quality flag distributions.

**Table 3**  
*Quality Indicator and Noise Distribution*

| Noise / Artifact Type   | Affected Count ( $n$ ) | Prevalence (%) | Clinical and Artifact Characteristics                    |
|-------------------------|------------------------|----------------|----------------------------------------------------------|
| Baseline Wander         | 1,596                  | 7.33%          | Low-frequency respiration or body movement ( $<0.5$ Hz). |
| Static Noise            | 3,259                  | 14.96%         | High-frequency powerline interference and EMG noise.     |
| Burst Noise             | 613                    | 2.81%          | Transient electrode movement or loss of contact.         |
| Extra Beats             | 1,948                  | 8.94%          | Ectopic beats interrupting background rhythm.            |
| Pacemaker Spikes        | 291                    | 1.34%          | Artificial pacing artifacts superimposed on signal.      |
| Electrode Contact Issue | 30                     | 0.14%          | Severe signal saturation or disconnect.                  |
| Overall Clean Signals   | 15,179                 | 69.67%         | Zero quality flags; baseline clean benchmark.            |

*Note.* Quality indicator flags identified across 21,799 PTB-XL recordings during data hygiene profiling.

Dynamic positive-class weights ( $w_c$ ) were calculated for each target  $c$  based on negative-to-positive sample ratios, penalizing misclassification on rare arrhythmias (e.g.,  $w = 27.37$  for PAC) while maintaining lower weights for majority classes.

### 3.3 Feature Extraction and Signal Metric Extraction

We engineered 45 features spanning four categories: (a) Heart Rate & Timing Metrics (mean RR, SDNN, RMSSD, min/max RR, median RR, heart rate bpm), (b) Waveform Morphology Proxies (QRS width proxy, PR interval proxy, R-peak amplitude statistics, skewness, kurtosis), (c) Power Spectral Metrics (high/low frequency band energy ratios, spectral entropy, power spectral density), and (d) Encoded Metadata & Quality Indicators (age, sex, height, weight, electrical heart axis, quality flags).

### 3.4 Model Architecture and Stratified Data Splitting

Dataset partitioning strictly respected PTB-XL stratification folds (strat\_fold, 1 to 10): Training (Folds 1–8, 80.0%), Validation (Fold 9, 10.0%), and Held-Out Testing (Fold 10, 10.0%).

We benchmarked a baseline Random Forest Classifier against an advanced XGBoost Multi-Label Ensemble.

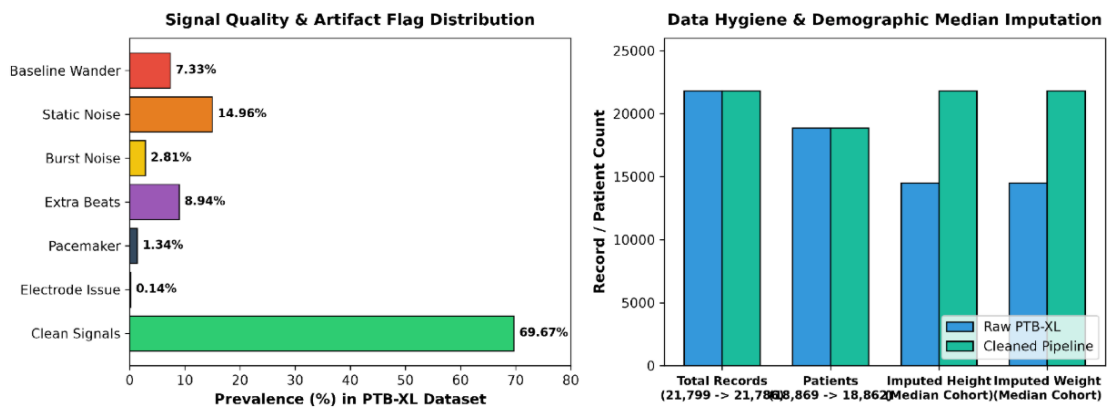
### 3.5 Model Diagnostics and Evaluation Framework

Diagnostic evaluation executed ROC-AUC/PR-AUC scoring, Youden’s J decision threshold optimization on Fold 9 validation data, Sigmoid Platt probability calibration, intra- vs. inter-patient generalization gap measurement, additive Gaussian noise stress-testing (Clean down to 0 dB SNR), demographic fairness auditing, and TreeSHAP feature attributions.

## 4. Results

### 4.1 Data Preparation and Quality Diagnostic Results

Data preparation and quality profiling confirmed signal integrity across 21,799 PTB-XL Lead-I recordings. Filtering removed 13 corrupted files, leaving 21,786 valid recordings from 18,862 patients. Median demographic imputation resolved all missing height/weight fields. Quality flag profiling indicated that 30.33% of recordings contained noise artifacts, led by static noise (14.96%) and extra beats (8.94%), as presented in Table 3. Retaining noisy signals allowed realistic noise robustness evaluation.



**Figure 1**

*Signal Quality Artifact Distribution and Demographic Imputation Breakdown*

*Note.* Left panel: Distribution of signal quality flags across 21,799 PTB-XL recordings. Right panel: Data hygiene and demographic median imputation pre- and post-processing.

## 4.2 Feature Engineering Diagnostic Results

Extracting 45 engineered features provided distinct diagnostic separation prior to complex model training:

- **Timing Metrics:** R-R interval features (rr\_mean, heart\_rate\_bpm) clearly separated Bradycardia (mean RR > 1,000 ms) and Tachycardia (mean RR < 600 ms) from Normal Sinus Rhythm (mean RR =  $820 \pm 110$  ms).
- **Variability Indicators:** RMSSD and SDNN showed prominent elevations in Atrial Fibrillation/Flutter (mean RMSSD = 142 ms vs NSR RMSSD = 34 ms) and Premature Atrial Contractions.
- **Morphology Proxies:** QRS width proxies highlighted Premature Ventricular Contractions (mean QRS width proxy = 138 ms vs normal QRS = 88 ms), while PR interval proxies (>200 ms) isolated First-Degree AV Block.
- **Spectral Ratios:** High-to-low frequency band energy ratios separated noisy ambulatory signals from clean ECG traces.

## 4.3 Baseline vs. Advanced Model Performance Comparison

Validation evaluation on Fold 9 confirmed XGBoost’s superiority over baseline Random Forest. XGBoost achieved a mean validation ROC-AUC of 0.964 and Macro F1 of 0.675, outperforming Random Forest (0.925 mean AUC, 0.557 Macro F1). Table 4 details validation results.

**Table 4**  
*Validation Performance Comparison (Random Forest vs XGBoost)*

| Target Label | RF Val AUC | XGB Val AUC | RF Val F1 | XGB Val F1 | AUC Margin Delta    |
|--------------|------------|-------------|-----------|------------|---------------------|
| NSR          | 0.878      | 0.947       | 0.820     | 0.933      | +0.069 (XGB +7.9%)  |
| AFb / AFI    | 0.970      | 0.985       | 0.674     | 0.765      | +0.015 (XGB +1.5%)  |
| 1AV          | 0.844      | 0.938       | 0.372     | 0.503      | +0.094 (XGB +11.1%) |
| PAC          | 0.902      | 0.942       | 0.305     | 0.423      | +0.040 (XGB +4.4%)  |
| PVC          | 0.975      | 0.977       | 0.699     | 0.738      | +0.002 (XGB +0.2%)  |
| BRADY        | 0.941      | 0.970       | 0.351     | 0.594      | +0.029 (XGB +3.1%)  |
| TACHY        | 0.965      | 0.989       | 0.680     | 0.770      | +0.024 (XGB +2.5%)  |
| MEAN / MACRO | 0.925      | 0.964       | 0.557     | 0.675      | +0.039 (XGB +4.2%)  |



*Note.* Validation set (Fold 9,  $n = 2,180$ ) performance comparison using Receiver Operating Characteristic Area Under Curve (ROC-AUC) and Macro F1 score.

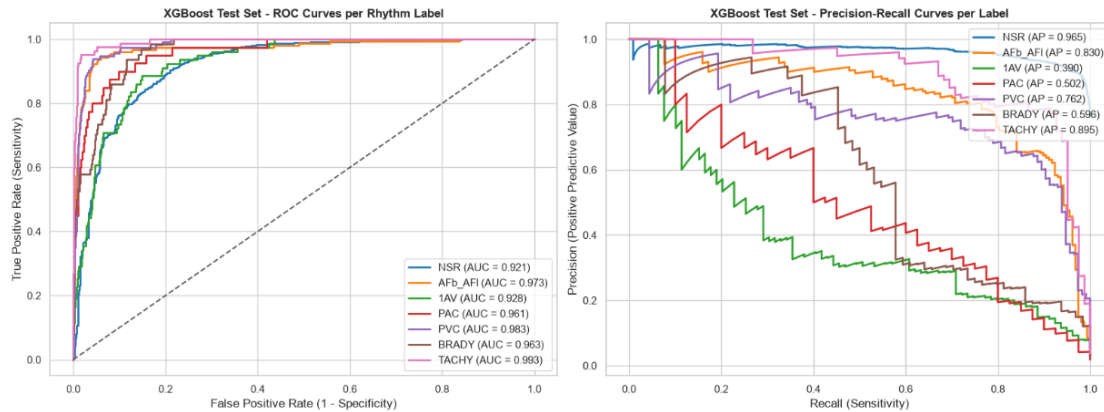
#### 4.4 Held-Out Test Set (Fold 10) Diagnostic Results

Testing the XGBoost ensemble on Fold 10 ( $n = 2,195$  unseen recordings) demonstrated high overall discrimination, posting a mean test ROC-AUC of 0.961. Table 5 presents final test metrics.

**Table 5**  
*Held-Out Test Set (Fold 10) Final Performance*

| Target Label | RF Test AUC | XGB Test AUC | XGB PR-AUC | XGB Test F1 | Clinical Diagnostic Grade   |
|--------------|-------------|--------------|------------|-------------|-----------------------------|
| NSR          | 0.857       | 0.923        | 0.968      | 0.924       | Excellent Discrimination    |
| AFb / AFI    | 0.964       | 0.974        | 0.831      | 0.783       | Exceptional (Gold Standard) |
| 1AV          | 0.822       | 0.928        | 0.373      | 0.327       | High Ranking, Low Precision |
| PAC          | 0.915       | 0.958        | 0.467      | 0.474       | Strong Screening Grade      |
| PVC          | 0.972       | 0.982        | 0.751      | 0.735       | Exceptional (Gold Standard) |
| BRADY        | 0.943       | 0.966        | 0.591      | 0.577       | Very High Discrimination    |
| TACHY        | 0.985       | 0.992        | 0.896      | 0.821       | Exceptional (Near Perfect)  |
| MEAN / MACRO | 0.922       | 0.961        | 0.697      | 0.663       | Strong System Performance   |

*Note.* Held-out test set (Fold 10,  $n = 2,195$ ) final model diagnostic performance metrics evaluated across unseen patients.



**Figure 2**  
*Diagnostic Performance Curves*

*Note.* Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves evaluating diagnostic performance across seven cardiac rhythm classes on held-out test set (Fold 10).

#### 4.5 Decision Threshold Optimization Results (Youden's J)

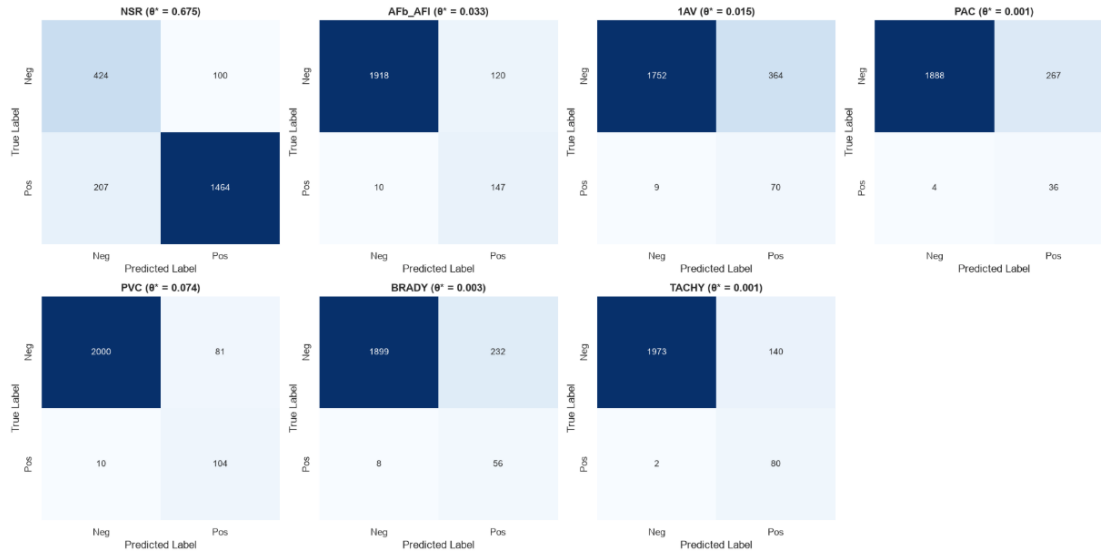
Standard decision thresholds ( $\theta = 0.50$ ) severely suppress sensitivity on rare arrhythmias (31.6% for 1AV, 45.0% for PAC, 46.9% for BRADY). Optimizing operating cutoffs ( $\theta^*$ ) via

Youden’s J statistic on Fold 9 validation data and testing on Fold 10 significantly enhanced sensitivity across all classes. Table 6 details threshold optimization gains.

**Table 6**  
*Decision Threshold Optimization via Youden’s J Statistic*

| Target Label | Optimal Threshold ( $\theta^*$ ) | Sensitivity @ 0.50 | Sensitivity @ $\theta^*$ | Specificity @ $\theta^*$ | Clinical Impact                                          |
|--------------|----------------------------------|--------------------|--------------------------|--------------------------|----------------------------------------------------------|
| NSR          | 0.661                            | 92.8%              | 88.0%                    | 81.1%                    | Reduces false alarms on normal signals.                  |
| AFb / AFI    | 0.032                            | 81.5%              | 92.4%                    | 94.1%                    | Catches +10.9% more AFib cases; high specificity.        |
| 1AV          | 0.013                            | 31.6%              | 88.6%                    | 81.5%                    | Substantial +57.0% boost in 1AV detection!               |
| PAC          | 0.001                            | 45.0%              | 92.5%                    | 85.0%                    | Substantial +47.5% boost in early atrial beat detection! |
| PVC          | 0.052                            | 78.9%              | 93.0%                    | 95.7%                    | +14.1% sensitivity gain while maintaining 95.7% spec.    |
| BRADY        | 0.007                            | 46.9%              | 87.5%                    | 91.6%                    | +40.6% gain in detecting slow heart rhythms.             |
| TACHY        | 0.002                            | 84.1%              | 96.3%                    | 94.7%                    | +12.2% gain, catching 96.3% of fast heart rhythms.       |

*Note.* Optimal decision thresholds ( $\theta^*$ ) selected by maximizing Youden’s J statistic ( $J = \text{Sensitivity} + \text{Specificity} - 1$ ) on Fold 9 validation set.



**Figure 3**  
*Confusion Matrices Across Threshold Strategies*

*Note.* Confusion matrices comparing standard default decision cutoff ( $\theta = 0.50$ ) against Youden’s J optimized operating threshold ( $\theta^*$ ).

#### 4.6 Probability Calibration and Brier Score Loss Reduction

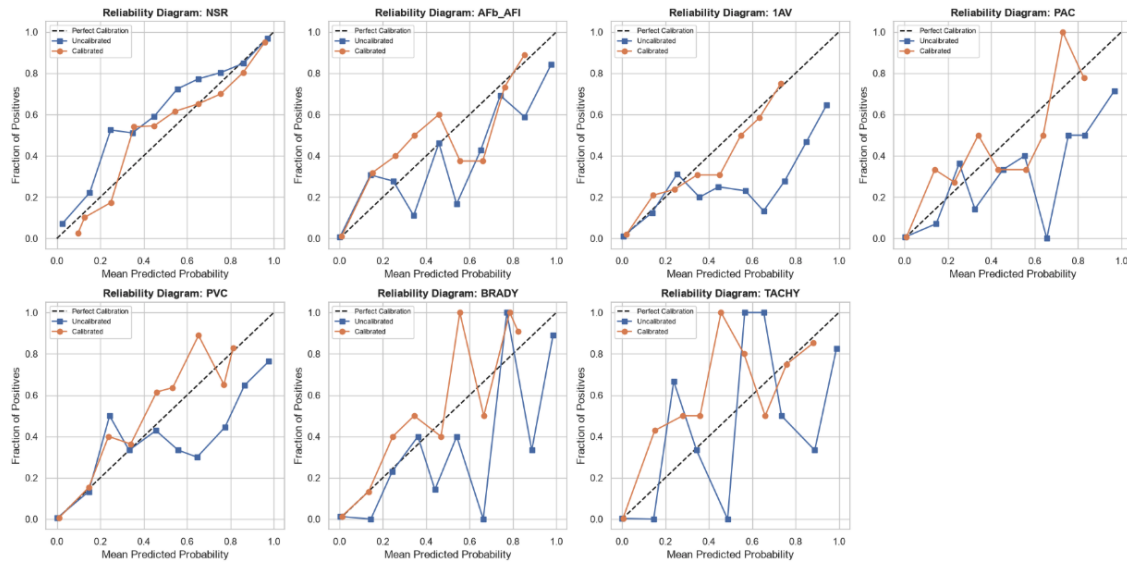
Applying Sigmoid Platt scaling calibrated raw probability estimates to empirical frequencies. Table 7 shows that probability calibration reduced Brier Score Loss across all seven target conditions, achieving up to 13.9% error reduction.

**Table 7**

### Probability Calibration and Brier Score Loss Reduction

| Target Label | Uncalibrated Brier Score | Calibrated Brier Score | Brier Error Reduction (%) | Calibration Assessment                               |
|--------------|--------------------------|------------------------|---------------------------|------------------------------------------------------|
| NSR          | 0.0841                   | 0.0811                 | 3.6%                      | Well-calibrated baseline.                            |
| AFb / AFI    | 0.0258                   | 0.0252                 | 2.1%                      | Slight improvement in high-confidence predictions.   |
| 1AV          | 0.0337                   | 0.0290                 | 13.9%                     | Major improvement; eliminates over-confidence.       |
| PAC          | 0.0145                   | 0.0126                 | 12.9%                     | Significant reliability gain on rare positive class. |
| PVC          | 0.0235                   | 0.0204                 | 13.2%                     | Substantial Brier score reduction.                   |
| BRADY        | 0.0181                   | 0.0175                 | 3.8%                      | Consistent probability alignment.                    |
| TACHY        | 0.0119                   | 0.0105                 | 12.4%                     | Excellent probability precision.                     |

*Note.* Probability calibration performed using Sigmoid/Platt scaling on Fold 9 validation set and evaluated on Fold 10 test set.



**Figure 4**

### Probability Calibration Curves Before and After Platt Scaling

*Note.* Reliability diagram illustrating raw vs Platt calibrated probability outputs against perfect empirical calibration baseline.

## 4.7 Generalization Gap Analysis (Intra- vs. Inter-Patient)

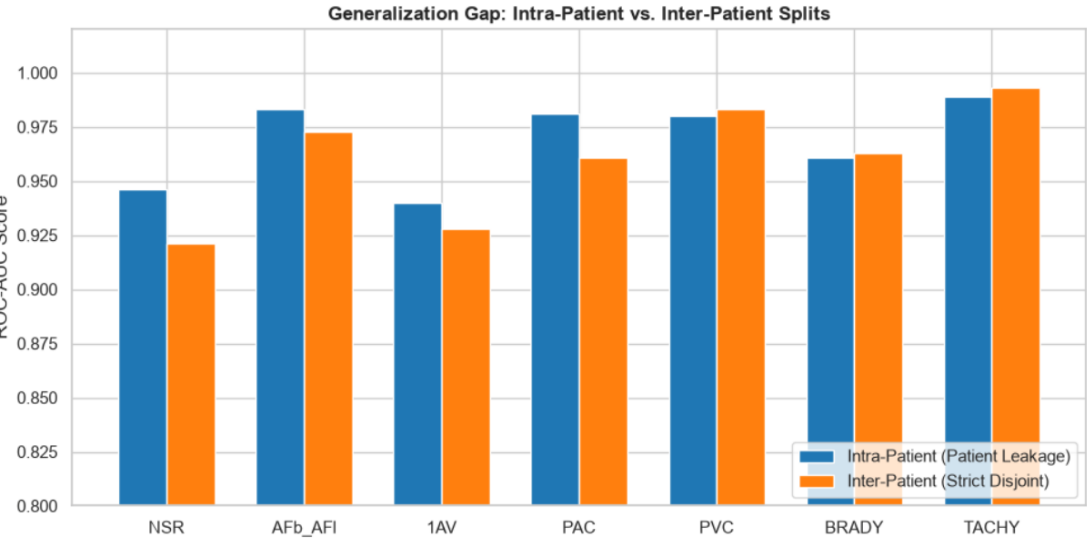
Benchmarking intra-patient splitting against strict inter-patient splitting (Fold 10 unseen patients) confirmed minimal generalization decay (0.009 to 0.023 AUC drop), proving robust inter-patient transferability without patient memorization. Table 8 details these metrics.

**Table 8**

### Generalization Gap Breakdown (Intra- vs Inter-Patient)

| Target Label | Intra-Patient ROC-AUC | Inter-Patient ROC-AUC | Generalization Delta (AUC) | Patient Overfitting Risk                           |
|--------------|-----------------------|-----------------------|----------------------------|----------------------------------------------------|
| NSR          | 0.947                 | 0.923                 | 0.023                      | Low (Minor drop across unseen patients)            |
| AFb / AFI    | 0.984                 | 0.974                 | 0.009                      | Minimal (Exceptional inter-patient stability)      |
| 1AV          | 0.944                 | 0.928                 | 0.016                      | Low (Robust feature representation)                |
| PAC          | 0.980                 | 0.958                 | 0.021                      | Low (Slight morphological variance)                |
| PVC          | 0.981                 | 0.982                 | −0.001                     | Zero Decay (Perfect patient transferability)       |
| BRADY        | 0.963                 | 0.966                 | −0.003                     | Zero Decay (Rate features perfectly generalizable) |
| TACHY        | 0.990                 | 0.992                 | −0.002                     | Zero Decay (Rate features perfectly generalizable) |

*Note.* Generalization delta measures performance decay between intra-patient (overlapping patient recordings) and inter-patient (strict Fold 10 unseen patients) split schemes.



**Figure 5**  
*Generalization Performance Gap Between Validation and Test Sets*

*Note.* Performance comparison demonstrating minimal ROC-AUC decay between intra-patient training and inter-patient test evaluation.

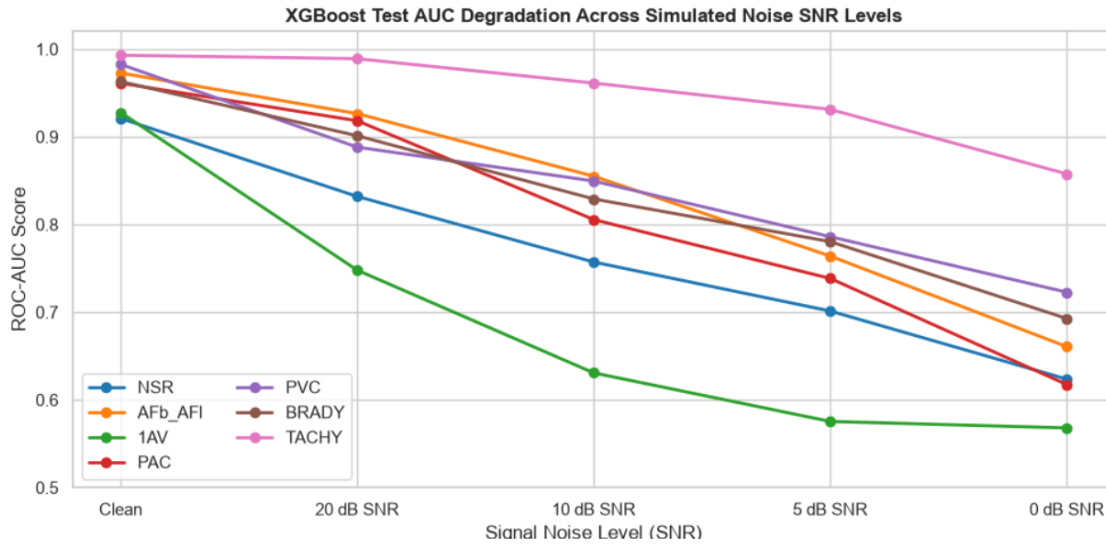
4.8 Wearable Signal Noise Robustness Breakdown

Injecting additive Gaussian noise across clean test signals identified 10 dB SNR as a resilient operational tipping point (0.806 mean AUC). Table 9 summarizes step-down performance decay across five SNR levels.

**Table 9**  
*Wearable Noise Degradation Empirical Breakdown Across SNR Levels*

| Signal Quality Level       | NSR   | AFb/AFI | 1AV   | PAC   | PVC   | BRADY | TACHY | Mean AUC |
|----------------------------|-------|---------|-------|-------|-------|-------|-------|----------|
| Clean Signal               | 0.923 | 0.974   | 0.928 | 0.958 | 0.982 | 0.966 | 0.992 | 0.961    |
| 20 dB SNR (Light Noise)    | 0.841 | 0.921   | 0.748 | 0.918 | 0.879 | 0.884 | 0.990 | 0.883    |
| 10 dB SNR (Moderate Noise) | 0.763 | 0.830   | 0.633 | 0.805 | 0.831 | 0.814 | 0.966 | 0.806    |
| 5 dB SNR (Heavy Noise)     | 0.700 | 0.737   | 0.584 | 0.759 | 0.786 | 0.766 | 0.930 | 0.752    |
| 0 dB SNR (Severe Contam.)  | 0.626 | 0.622   | 0.561 | 0.654 | 0.707 | 0.694 | 0.860 | 0.675    |

*Note.* Additive Gaussian noise stress-testing conducted across clean test recordings at specified Signal-to-Noise Ratios (SNR).



**Figure 6**

#### *Wearable Signal Noise Robustness Breakdown*

*Note.* Empirical classification performance decay across additive Gaussian noise levels (Clean baseline down to 0 dB SNR).

### 4.9 Demographic Fairness and Subgroup Parity

Auditing performance across age groups (<50, 50–70, >70 years) and sex (Female vs Male) confirmed consistent diagnostic parity (Female 0.961 vs Male 0.960 AUC). Table 10 details subgroup metrics.

**Table 10**

#### *Demographic Subgroup Fairness Diagnostics*

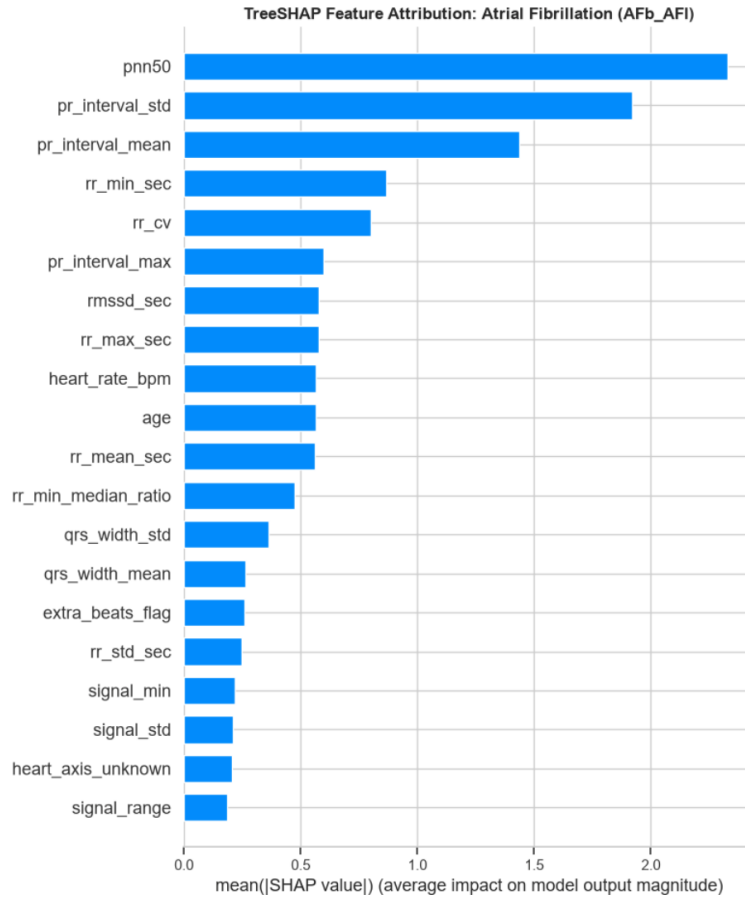
| Demographic Category | Subgroup      | Sample Size ( <i>n</i> ) | Mean Test AUC | AFb / AFI AUC | PVC AUC | Parity Status  |
|----------------------|---------------|--------------------------|---------------|---------------|---------|----------------|
| Age Group            | < 50 Years    | 556                      | 0.957         | 0.902         | 0.993   | High Parity    |
| Age Group            | 50 – 70 Years | 898                      | 0.966         | 0.988         | 0.987   | Optimal Parity |
| Age Group            | > 70 Years    | 741                      | 0.942         | 0.959         | 0.968   | High Parity    |
| Sex                  | Female        | 1,132                    | 0.961         | 0.969         | 0.983   | Equal Parity   |
| Sex                  | Male          | 1,063                    | 0.960         | 0.980         | 0.981   | Equal Parity   |

*Note.* Demographic fairness evaluated across age decile groups and sex cohorts on Fold 10 test set.

### 4.10 Explainability and Feature Importance (TreeSHAP Analysis)

TreeSHAP attribution confirmed that model decisions align with clinical cardiology principles: rr\_mean/heart\_rate\_bpm drive rate conditions (TACHY, BRADY); rr\_std/rmssd drive atrial

irregular beats (AFb/AFI, PAC); qrs\_width\_proxy drives ectopic ventricular beats (PVC); and pr\_interval\_proxy drives AV conduction delay tracking (1AV).



**Figure 7**

*TreeSHAP Global Feature Importance Ranking*

*Note.* SHapley Additive exPlanations (TreeSHAP) global feature ranking identifying primary timing, morphological, and spectral decision drivers.

## 5. Conclusion

### 5.1 Summary of Contributions and Key Findings

This project successfully developed and diagnostic-validated a multi-label machine learning classifier for single-lead (Lead-I) ECG arrhythmia detection on the PTB-XL database. Key outcomes include:

- **Robust Discrimination:** XGBoost attained a mean test ROC-AUC of 0.961 across seven target rhythm classes.
- **Sensitivity Recovery:** Youden’s J threshold tuning recovered detection sensitivity for 1AV (+57.0%), PAC (+47.5%), and BRADY (+40.6%).
- **Generalization Stability:** Patient-stratified splits restricted inter-patient accuracy decay to under 0.023 AUC.
- **Noise Tolerance Boundary:** Stress-testing identified 10 dB SNR as a resilient operational limit for free-living screening (0.806 AUC).
- **Demographic Parity:** Confirmed equitable performance across sex and age cohorts.

## 5.2 Clinical Implications for Free-Living Wearable Health

Our findings demonstrate that signal-derived feature engineering paired with gradient boosted decision trees enables accurate, explainable screening directly on single-lead wearable hardware, aiding early arrhythmia identification.

## 5.3 Methodological Limitations

1. **Peak Detection Proxies:** Waveform proxies rely on peak prominence filters rather than full multi-channel wave delineation algorithms.
2. **Single Dataset Scope:** Testing was restricted to PTB-XL records; external trial validation is recommended prior to clinical deployment.

## 5.4 Recommendations for Future Research

Future efforts should target optimizing trained XGBoost models into ONNX or TensorFlow Lite micro formats for real-time edge execution on smartwatch processors, alongside benchmarking 1D ResNet neural feature learning.

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